

The background is a blurred image of a computer monitor. The screen shows a dark-themed code editor with some text in blue and green. At the bottom, a Windows taskbar is visible with various application icons like Chrome, Photoshop, and others.

Introduction To Programming Using C++ Level 2

Call ByRefByVal Important Review

The background of the slide is a photograph of a lecture hall. In the foreground, the silhouettes of several students are visible, some sitting and some standing. In the background, a large chalkboard is covered with faint, illegible lines of text, suggesting a lecture or a review session. The overall lighting is dim, with the chalkboard being the primary light source.

Introduction To Programming Using C++

Level 2

Call ByRefByVal Important Review

Definition

Call by Value

-  Call by Value means passing a copy of the argument's value to the function.

-  The function receives its own parameter variable, so changing the parameter does not change the original variable in the calling function.

Call by Reference

-  Call by Reference means passing a reference to the original variable to the function.

-  The function parameter becomes another name (Alias) for the original variable, so both refer to the same variable.

Explanation

Call by Value

-  When a variable is passed by value, the function receives a separate copy of its value.

-  Any modification made to the parameter affects only the copy.

-  The original variable remains unchanged.

Call by Reference

-  When a variable is passed by reference using &, the function works directly with the original variable.

-  No separate copy of the variable is created.

-  Any modification made to the reference also modifies the original variable.

Procedures

1. Create a variable in main().
2. Pass the variable to a function.
3. For Call by Value, define the parameter normally:
 void Function1(int x)
4. For Call by Reference, use &:
 void Function1(int& x)
5. Modify the parameter inside the function.
6. Return to main().
7. Check the value of the original variable.

💡 Example

◆ Call by Value

// Function to demonstrate Call by Value

```
void Function1(int x)
```

```
{
```

```
    x++;
```

```
}
```

```
int main()
```

```
{
```

```
    int a = 10;
```

```
    Function1(a);
```

```
    cout << "\na after calling Function1 = " << a << endl;
```

```
    return 0;
```

```
}
```

➡ Result

a after calling Function1 = 10

Note

◆ The & in:

```
void Function1(int& x)
```

means that x is a reference parameter.

◆ A reference acts as another name (Alias) for the original variable.

◆ You can use &Name with cout to print the memory address of a variable:

```
cout << &Name << endl;
```

◆ The address is normally displayed in hexadecimal format.

⊘ Warning

- ◆ Do not confuse Call by Reference with passing a pointer.
- ◆ In C++, references and pointers are different concepts, even though both can be used to allow a function to modify the original object.
- ◆ Call by Value does not necessarily mean that every argument always requires a large physical memory copy. The important concept is that the function receives a separate value rather than a reference to the original object.



Comparison

▨ Call by Value

- ✓ A copy of the value is passed to the function.
- ✓ The parameter is independent from the original variable.
- ✓ Changes made inside the function do not affect the original variable.

▤ Call by Reference

- ✓ A reference to the original variable is passed to the function.
- ✓ The parameter refers to the same variable as the original.
- ✓ Changes made inside the function affect the original variable.

✦✦ Summary

- ✓ Call by Value → The function receives a separate value, so changes do not affect the original variable.
- ✓ Call by Reference → The function works with the original variable through a reference, so changes affect the original variable.

Thank you 😊